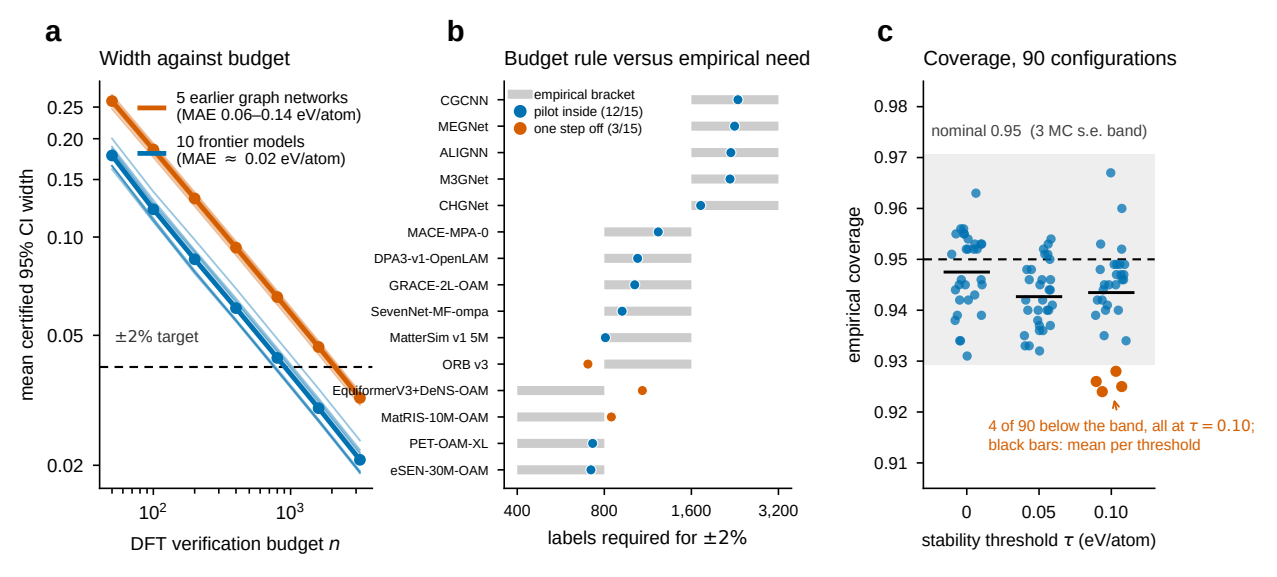
a
Width against budget
5 earlier graph networks
(MAE 0.06–0.14 eV/atom)
10 frontier models
(MAE ≈ 0.02 eV/atom)
±2% target
mean certified 95% CI width
0.25
0.20
0.15
0.10
0.05
0.02
10^2
10^3
DFT verification budget n
b
Budget rule versus empirical need
empirical bracket
pilot inside (12/15)
one step off (3/15)
CGCNN
MEGNet
ALIGNN
M3GNet
CHGNet
MACE-MPA-0
DPA3-v1-OpenLAM
GRACE-2L-OAM
SevenNet-MF-ompa
MatterSim v1 5M
ORB v3
EquiformerV3+DeNS-OAM
MatRIS-10M-OAM
PET-OAM-XL
eSEN-30M-OAM
400
800
1,600
3,200
labels required for ±2%
c
Coverage, 90 configurations
nominal 0.95 (3 MC s.e. band)
empirical coverage
0.98
0.97
0.96
0.95
0.94
0.93
0.92
0.91
4 of 90 below the band, all at τ = 0.10;
black bars: mean per threshold
0
0.05
0.10
stability threshold τ (eV/atom)